

```

13.     for i in range(numberOfBirthdays):
14.         # The year is unimportant for our simulation, as long as all
15.         # birthdays have the same year.
16.         startOfYear = datetime.date(2001, 1, 1)
17.
18.         # Get a random day into the year:
19.         randomNumberOfDay = datetime.timedelta(random.randint(0, 364))
20.         birthday = startOfYear + randomNumberOfDay
21.         birthdays.append(birthday)
22.     return birthdays
23.
24.
25. def getMatch(birthdays):
26.     """Returns the date object of a birthday that occurs more than once
27.     in the birthdays list."""
28.     if len(birthdays) == len(set(birthdays)):
29.         return None # All birthdays are unique, so return None.
30.
31.     # Compare each birthday to every other birthday:
32.     for a, birthdayA in enumerate(birthdays):
33.         for b, birthdayB in enumerate(birthdays[a + 1 :]):
34.             if birthdayA == birthdayB:
35.                 return birthdayA # Return the matching birthday.
36.
37.
38. # Display the intro:
39. print('''Birthday Paradox, by Al Sweigart al@inventwithpython.com
40.
41. The Birthday Paradox shows us that in a group of N people, the odds
42. that two of them have matching birthdays is surprisingly large.
43. This program does a Monte Carlo simulation (that is, repeated random
44. simulations) to explore this concept.
45.
46. (It's not actually a paradox, it's just a surprising result.)
47. ''')
48.
49. # Set up a tuple of month names in order:
50. MONTHS = ('Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
51.           'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec')
52.
53. while True: # Keep asking until the user enters a valid amount.
54.     print('How many birthdays shall I generate? (Max 100)')
55.     response = input('> ')
56.     if response.isdecimal() and (0 < int(response) <= 100):
57.         numBDays = int(response)
58.         break # User has entered a valid amount.
59. print()
60.
61. # Generate and display the birthdays:
62. print('Here are', numBDays, 'birthdays:')
63. birthdays = getBirthdays(numBDays)
64. for i, birthday in enumerate(birthdays):
65.     if i != 0:
66.         # Display a comma for each birthday after the first birthday.
67.         print(', ', end='')

```